

A PLAIN-LANGUAGE GUIDE

Health Services Research

How epidemiologic and statistical methods are used to study who gets care, how well the system delivers it, and what it ultimately achieves for health.

Based on “Health Services Research” by Thomas Schäfer, Christian A. Gericke, and Reinhard Busse — Handbook of Epidemiology

ROADMAP

Eight Stops

- 01** What health services research is, the inputs-to-outcomes model of care, and the levels at which it is studied
- 02** The study designs it relies on, and the complex models its longitudinal data demand
- 03** Where its data come from, the measurement error they carry, and how confounding is controlled
- 04** Demand versus need, assessing what a population requires, and the barriers that limit access
- 05** How systems raise and allocate money, weigh spending choices, and pay providers
- 06** Judging the process of care itself — effectiveness, appropriateness, quality, and public rankings
- 07** Measuring health outcomes, patient experience, and efficiency — and comparing them across countries
- 08** Bringing it all back together

FOUNDATIONS

A Field Built From Many Disciplines

The Institute of Medicine (1994) defines health services research as the multidisciplinary field that studies how social factors, financing systems, organizational structures and processes, health technologies, and personal behavior affect access to care, its cost and quality, and ultimately health and well-being.

Last's dictionary of epidemiology gives a similar definition: the integration of epidemiologic, sociologic, economic, and other analytic sciences in the study of health services — usually in relation to health-related outcomes.

It draws heavily on epidemiologic methods, but points them at the health care system itself rather than at a disease

Its goal is unbiased scientific evidence that can influence health policy — how care is organized, delivered, financed, and paid for

FOUNDATIONS

Three Basic Dimensions of Care

Nearly every question health services research asks falls under one of three basic dimensions of care.

Deciding what to provide

Which services, to whom, and under what conditions the system should make available

Providing it well

Whether care, once offered, is delivered effectively, appropriately, and to a consistent standard of quality

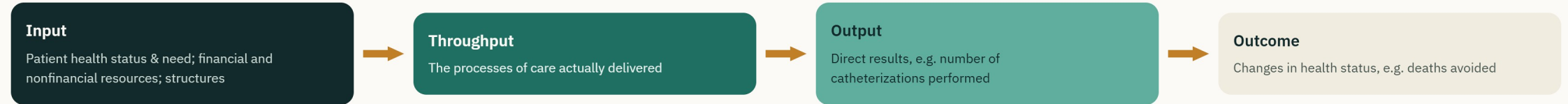
Outcomes

Whether all of it actually improves health, function, and well-being in the end

This deck follows that same logic — from the inputs and structures that shape care, through the process of delivering it, to the outcomes it produces.

FOUNDATIONS

Care as a System: Input, Throughput, Output, Outcome



Feedback loops complicate this simple chain: the environment, other social systems, the educational system, the population's health, and the financial resources available all loop back to reshape the next round of input — part of why isolating the causal effect of any single reform is so difficult.

Author's illustration of the input-output model of health care (after Aday and colleagues), not a reproduction of the chapter's original figure.

FOUNDATIONS

Why Causal Inference Is Especially Hard Here

Health systems rarely hold still long enough for a clean before-and-after comparison.

Piecemeal reform

Health policy reforms are rarely introduced as one clean, isolated change — several things shift at once, confounding any single reform’s estimated effect

A moving target

By the time an evaluation is designed and run, the system it describes may already have changed again

Researchers sometimes describe this as the “before the intervention is after the intervention” dilemma: the baseline period meant to represent life before a reform may already reflect anticipation of it, or overlap with other ongoing changes — making a truly clean comparison hard to find.

FOUNDATIONS

Three Levels of Analysis

Macrolevel

The health system as a whole, or national policy — financing structure, coverage, regulation

Mesolevel

Interorganizational relationships — between payers and providers, or across institutions

Microlevel

Individual patient care, and specific technologies or treatments

The same research question can often be studied at more than one level — and the data sources available differ sharply by level, as the next slide shows.

FOUNDATIONS

Matching Data Sources to the Level of Analysis

Data Source	Population	System / Institution	Patient
Census & vital statistics	Yes	—	—
Patient records	—	Yes	Yes
Insurance records & claims	Yes	Yes	Yes
Qualitative studies	Varies	Varies	Varies

After Aday and colleagues (1998): the same data source can often serve more than one level of analysis, but some — like insurance claims — are unusually versatile, which is part of why administrative data have become so central to the field.

STUDY DESIGNS

The Same Toolkit as Epidemiology, Aimed at the System

Health services research borrows its study designs directly from epidemiology — randomized trials, cohorts, case-control studies, cross-sectional surveys — but points them at exposures like insurance status, delivery-system type, or a quality-improvement program, and outcomes like cost, access, or patient-reported quality of life.

Each design carries its familiar epidemiologic trade-offs between internal validity, external validity, cost, and speed — but HSR's typical data source, administrative claims, also reshapes how those designs are actually carried out.

STUDY DESIGNS, 1 OF 6

Randomized Controlled Trials

Still the gold standard for internal validity — randomization balances both known and unknown confounders between groups. But a trial’s controlled, closely monitored setting often gives it low external validity: results measure *efficacy*, care under ideal conditions, which can differ sharply from *effectiveness*, how well an intervention performs in everyday practice.

Lipid Research Clinics trial

Tested cholesterol-lowering drug therapy for reducing coronary heart disease

Women’s Health Initiative

Large trial evaluating hormone therapy and other preventive interventions in postmenopausal women

RAND Health Insurance Experiment

Randomized different cost-sharing plans to measure their effect on utilization and health

Observational Studies: Typical Exposures & Outcomes

Typical exposures

Conditions of inequality; health states or needs; medical interventions; delivery-system type (HMO, PPO, fee-for-service); quality-improvement programs; cost-containment programs

Typical outcomes

Access or vaccination rates; health status via tracer diagnoses; life-years gained; patient-reported outcomes; quality scores (e.g. HEDIS); appropriateness of care; cost

Because randomizing people to an insurance type or a delivery system is often impossible or unethical, most HSR questions about how the system itself is organized are studied observationally — making confounding control especially important (covered later in this section).

STUDY DESIGNS, 3 OF 6

Case-Control Studies

Low-cost and efficient for rare outcomes, case-control designs are a common choice when HSR needs an answer quickly and cheaply.

Vaccine efficacy

Comparing vaccination history between disease cases and matched controls

Drug adverse effects

Comparing prior drug exposure between people with and without a suspected adverse event

Screening programs

Evaluating whether prior screening reduces the odds of advanced disease at diagnosis

STUDY DESIGNS, 4 OF 6

Cohort Studies: Rarer Here Than in Chronic Disease Epidemiology

Health systems, unlike biology, keep changing — a delivery system or payment policy followed for years may no longer resemble what it was at baseline, which limits how often true cohort designs get used. Most HSR cohorts today are built from administrative data rather than purpose-recruited participants.

HMO hip-fracture cohort

Followed patients after hip fracture surgery to compare 30-day and 5-year mortality across providers

The German National Cohort (NAKO)

A large population-based cohort recruiting since 2014 across 18 study centers — a rare example of a purpose-built cohort at national scale

STUDY DESIGNS, 5 OF 6

Cross-Sectional Studies

A snapshot of prevalence is often exactly what health planning needs — cross-sectional surveys are a mainstay of needs assessment, and of the “variance in practice” approach that compares how care is actually delivered across areas or providers at a single point in time.

Worked example

A US study of reproductive-age women found menstrual disorders, adnexal conditions, and uterine fibroids to be the most common gynecologic conditions reported — the kind of prevalence estimate that shapes service planning.

The limitation is structural: a cross-sectional design cannot establish temporal sequence between exposure and outcome, and it systematically overrepresents conditions of long duration simply because they are more likely to be “caught” by a single survey.

Claims Data Need More Complex Models

Administrative claims are naturally longitudinal — the same patient generates repeated records over time, with exposures that can change mid-study and outcomes that recur. Two design responses stand out:

Case-crossover designs

Each patient serves as their own control, comparing exposure just before an event to exposure at other times — useful when claims carry little detail on disease severity

Multilevel regression

Patients are naturally clustered within physicians, within practices, within health plans — multilevel models are now a standard HSR tool for handling that nested correlation

DATA & ERROR

Surveys: Purpose-Built, But Not Free

Surveys are classified by their unit of observation, target population, type of data collected, and mode of access — in person, by mail, by telephone, or self-administered.

Germany

The EVaS Study (1980s); the National Health Interview and Examination Survey (GHS, 1997–1999 and 2008–2011); the Health Update telephone survey; KIDA; “Gesundheit in Deutschland aktuell”

United States

The Medicare Current Beneficiary Survey (MCBS) — a four-year rotating panel with three interviews per year; the Medical Expenditure Panel Survey (MEPS)

Computer-assisted telephone interviews (CATI) are a common mode: Germany’s 2002 National Telephone Health Survey reached about 8,000 respondents this way.

Official Statistics: The Fading Autopsy

Mortality and other vital statistics are a foundational HSR data source — but their quality has been quietly eroding as one of their key validators, the autopsy, disappears.

Germany, 2005–2014 Autopsy rates fell from 13.0% to 8.6% at university hospitals, and from 6.3% to 3.8% at community hospitals	The Netherlands, 1977–2011 Fell from 31.4% to just 7.7%	United States, today Now under 5%, part of a worldwide decline underway since the 1950s
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Fewer autopsies mean fewer independent checks on the cause-of-death data that mortality statistics — and much HSR built on top of them — ultimately rest on.

DATA & ERROR, 3 OF 8

Administrative Databases: Cheap, Huge, and Built for Billing

Abstracted from discharge records, prescriptions, and bills, administrative data are routinely collected, cover huge populations, and cost little to obtain — but they were designed for reimbursement, not research, and are often incomplete or miscoded as a result.

US Medicare & Medicaid

Confidentiality-protected, longitudinal, person-level databases that can track a beneficiary from age 65 until death

Germany, since 2004

Mandatory transfer of billing data to sickness funds created large anonymized and pseudonymized research databases

Data-reduction techniques — comorbidity indices and propensity scores among them — help condense these very large, messy databases into workable analytic variables.

DATA & ERROR, 4 OF 8

Why HSR Measurement Error Runs High

Administrative data carry error risks beyond those of a typical epidemiologic study:

Data not needed for reimbursement is often left outdated or unchecked

A diagnostic code on a claim may reflect a test ordered to *rule out* a disease, not a confirmed diagnosis

ICD coding schemes are revised over time and applied inconsistently for vague or psychiatric conditions

Because billing data shape payment, they are also vulnerable to distortions rarely seen in clinical research data: “upcoding” (billing a more severe diagnosis than is present), “coding proliferation” (adding codes to justify more services), and outright “gaming” of the reimbursement system.

Diagnoses Don’t Always Stick From Year to Year

A 1998 MedPAC study checked whether 11 serious diagnoses, once coded on a claim, reappeared on that same patient’s claims the following year. For most, the answer was barely better than a coin flip.

Condition	Diagnostic Persistence
Chronic obstructive pulmonary disease	62%
Congestive heart failure	61%
Hypertension	59%
Dementia	59%
Dialysis-dependent renal disease	59%
High-cost diabetes	58%
Renal failure	56%
Rheumatoid arthritis	55%
Coronary artery disease	53%
Quadriplegia / paraplegia	52%
Stroke	51%

MedPAC (1998). Persistence below 100% does not necessarily mean the original diagnosis was wrong — but it does mean a single year’s claims data will misclassify a meaningful share of patients on any of these conditions.

Sampling: Five Ways to Draw a Probability Sample

The overarching rule: a scientifically valid sample means **probability** sampling, where every unit's chance of selection is known and controlled. A volunteer sample, or one an investigator hand-picks by judgment, is not a valid substitute for inference to the wider population — the five specific designs below are all ways of achieving a genuine probability sample.

Simple random

Every unit has an equal, known chance of selection

Systematic

Every kth individual on a list is selected

Stratified

Sampling within homogeneous strata reduces error

Cluster

Cuts cost, but loses precision versus simple random

Multistage

Combines several of the above in stages

Clustered samples create an intraclass correlation (a “cluster effect”): people within the same cluster tend to resemble each other, which understates true variance if ignored. Standard software often gets this wrong — specialized packages like SUDAAN, or the survey modules built into SAS, Stata, and SPSS, are needed to estimate variance correctly.

Confounding, Renamed “Risk Adjustment”

HSR uses the same confounding-control logic as epidemiology, under its own name. With very large administrative databases, standardization and multivariate modeling are far more common than stratification. Typical confounders adjusted for: age, sex, ethnicity, income, and smoking.

Continuous outcomes Ordinary least squares regression — e.g. for log-transformed cost data	Count outcomes Poisson or negative binomial regression	Binary outcomes Binomial or logistic regression
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Comorbidity is handled either as a set of dummy variables for each condition, or as a single premodeled, aggregated comorbidity index — a data-reduction trick that keeps regression models manageable.

Instrumental Variables: Powerful, Rarely Used

An instrumental variable predicts exposure without directly affecting the outcome itself — letting researchers approximate randomization from observational data. The technique is powerful but demanding to apply well, so it remains uncommon in HSR.

Differential distance to hospitals

Newhouse and McClellan (1998) used a patient's relative distance to a catheterization-capable hospital as an instrument for studying catheterization's effect on mortality after a heart attack

Physician preference

Kollhorst and colleagues (2016) used a prescriber's usual preference between a traditional NSAID and a COX-2 inhibitor as an instrument, since that preference predicts the prescription but not the patient's underlying risk

DEMAND, NEED & ACCESS

Demand and Need Are Not the Same Thing

Demand, in the economic sense, is the quantity of care buyers wish to buy at a given price — and health functions as both a consumption good (valued for itself) and a capital good (an investment that pays off later).

Need is a distinct concept, further split into *felt* need (what a person believes they need), *expressed* need (demand that felt need translates into), and *normative* need (what an expert judges to be required). A related idea, “capacity to benefit” (Williams and Culyer), reframes need as the ability to actually gain health from a service — not just wanting or lacking it.

Three Ways to Assess a Population's Needs

Epidemiologically based

Combines disease incidence and prevalence with evidence on effective interventions

Comparative

Compares service levels or outcomes across similar populations to spot shortfalls

Corporate

Draws on the knowledge, experience, and priorities of local stakeholders and professionals

None of the three is sufficient alone — good needs assessments typically triangulate across all three.

DEMAND, NEED & ACCESS, 2 OF 6

Worked Example: Carotid Endarterectomy in England

An epidemiologically based needs assessment estimated the true need for carotid endarterectomy (a stroke-prevention surgery) at about 153 operations per million people per year.

153

Estimated need, per million people per year

35–89

Actual operation rates observed, per million per year

0.47

Overall use-to-need ratio (range 0.28–0.81 across districts)

Fewer than half of the operations judged necessary were actually being performed, with wide variation by district and further inequity by age and sex — exactly the kind of gap a needs assessment is designed to expose.

Utilization Varies for Reasons Beyond Need

Sex differences

Appendectomy rates differ by sex in ways that outstrip any plausible difference in true disease incidence — a sign that diagnostic and treatment thresholds themselves can vary by patient group

Clinician judgment

A classic comparison of Boston and New Haven found Boston spent \$889 per capita on hospital care to New Haven’s \$451, with no difference in mortality — leading researchers to conclude that Boston was simply overutilizing care

These “small-area variation” findings recur throughout HSR (see the Process of Care section) and are a central reason the field studies utilization independently from need.

A deeper limitation: utilization data only include people who already overcame whatever stood between them and care — distance, transport, fear, a waiting list. Observed use is therefore a *selected* view of access, not a direct measure of it — low utilization does not necessarily mean low need.

Geography and Money Both Shape Access

Geographic access

A UK study of coronary revascularization found large variation in access simply based on where a patient lived relative to specialist centers

The inverse care law

Julian Tudor Hart's influential 1971 observation: the availability of good medical care tends to vary *inversely* with the need for it in the population served

Financial access barriers work the same direction: out-of-pocket costs and cost-sharing tend to deter exactly the lower-income patients who often have the greatest underlying need — the subject of the next slide's famous experiment.

The RAND Health Insurance Experiment

This landmark randomized trial assigned families to insurance plans with different levels of cost-sharing, then tracked their use of care and their health.



For most people, cutting back on care under cost-sharing made no detectable difference to health. But the poorest, sickest 6% of participants showed real harm — elevated blood pressure, worse anemia, and worse vision — a finding still cited in debates over cost-sharing today.

Unmet Need and Universal Coverage

“Unmet need” — care a person recognized they needed but did not receive, whether from cost, distance, or waiting times — has become a standard access indicator. The EU’s Statistics on Income and Living Conditions (EU-SILC) survey tracks it across member states.

A comparability caution: unmet-need figures are sensitive to culture and to how a survey is worded — some ask only about examination or consultation, others about examination and treatment together — so a cross-country difference can reflect the questionnaire as much as the underlying need.

These same demand, need, and access concepts sit directly underneath the global push for Universal Health Coverage: a system can only claim to cover its population if unmet need — not just formal insurance status — is actually low.

FINANCING & ORGANIZATION

Where the Money Comes From, and Where It Goes

Financing can be studied from three angles: the ultimate sources of funding, a full National Health Accounts breakdown of flows through the system, or a focus on how resources are finally allocated to providers.

Beveridge-type (NHS)

Funded mainly through general taxation

Bismarck-type

Funded mainly through payroll-based social insurance contributions

International comparison is standardized through the System of Health Accounts (SHA) and its International Classification for Health Accounts (ICHA). In the US, 27.3 million people — 8.3% of the population — remained uninsured in 2021.

Public Priorities Don't Always Match Spending

A WHO-affiliated international comparison study surveyed roughly 48,000 respondents across 12 study areas, asking which health issues mattered most to them.

A striking mismatch emerged: the areas respondents rated as having the **highest** societal interest tended to have the **lowest** per-capita health expenditure actually devoted to them — a gap between what the public says it values and where health spending actually flows.

Four Ways to Weigh Spending Choices

Method	Compares Cost Against	Best Suited To
Cost-minimization	Nothing — assumes outcomes are equivalent	Choosing between treatments already known to work equally well
Cost-effectiveness	A natural clinical unit, e.g. life-years gained	Comparing options that share a common outcome measure
Cost-utility	A preference-weighted unit, e.g. QALYs	Comparing options across very different kinds of health benefit
Cost-benefit	A monetary value placed on the outcome itself	Comparing health spending against entirely different kinds of public investment

Whichever method is used, it should state whose costs are being counted — an insurer, a hospital, a patient, or society as a whole — since the answer can change with the perspective taken; a societal perspective is the most common default. And “doing nothing” is not a costless baseline: it is itself an alternative, with its own costs and consequences to weigh against every other option.

Three Ways to Elicit a Health-State Preference

A quality-adjusted life year (QALY) weights a year of life by the quality of health lived in it, on a scale from 0 (dead) to 1 (perfect health). That weight has to come from somewhere:

Standard gamble

Asks what risk of death someone would accept for a chance at perfect health, versus a certain intermediate state

Time trade-off

Asks how many years of full health someone would trade for more years in a poorer state — the easiest of the three to administer

Category rating

Asks someone to place a health state directly on a labeled numeric or visual scale

Average Cost-Effectiveness Can Mislead

Doubilet and colleagues (1986) described three scenarios for judging a new option against an existing one: it may cost less and work better (an easy yes), cost more and work worse (an easy no), or — the hard case — cost more *and* work better, which requires weighing the added cost against the added benefit.

Worked example: the sixth guaiac stool test for colorectal cancer screening

Averaged across everyone screened, six sequential tests cost about **\$2,451** per additional case detected. But the *incremental* cost of that sixth test alone — over and above the first five — was roughly **\$47 million** per additional case it caught.

The average cost-effectiveness ratio looked reasonable; the incremental ratio — the one that actually describes the decision at the margin — revealed the true cost of adding that sixth test.

The Fine Print of Economic Evaluation

Three cost types

Direct (treatment itself), indirect (lost productivity), and intangible (pain, suffering) costs all belong in a full accounting, alongside the opportunity cost of what else the money could have bought

Discounting

Future costs and benefits are discounted back to present value, typically at 3–5% a year, since a benefit received later is worth less than one received now

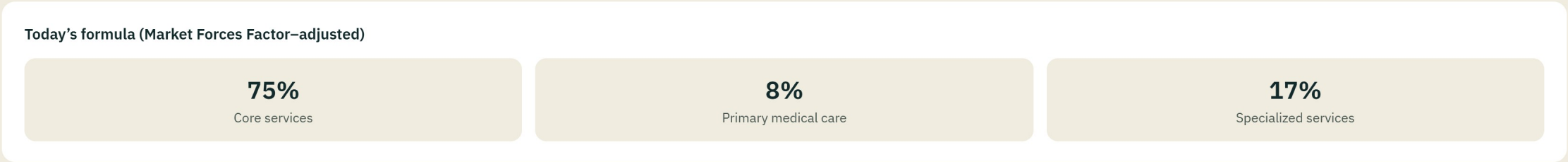
Sensitivity analysis

Key assumptions are varied to see how much the conclusion depends on them

Oregon’s attempt to rank Medicaid-covered services by cost-effectiveness ran into exactly this limit: a ranking based on cost-effectiveness alone left out equity and ethical considerations that mattered enormously to the public — a reminder that cost-effectiveness analysis informs a decision but cannot make it alone.

Weighted Capitation, NHS-Style

In tax-funded, Beveridge-type systems, money typically flows to local areas as a capitation payment — a fixed amount per resident, weighted for need. England’s formula has been revised repeatedly: the Resource Allocation Working Party (RAWP) formula from 1976, updated in 1995, followed by the Advisory Committee on Resource Allocation (ACRA) from 1999.



A real criticism: using historical utilization as a stand-in for need can simply reproduce whatever inequities already shaped that utilization. The chapter draws a sharp line between “equal access to health care” and the broader goal of “reducing health inequalities” — a formula can achieve the first while doing little for the second.

FINANCING & ORGANIZATION, 7 OF 11

Risk-Adjusted Capitation: Fixing “Cream-Skimming”

A flat, unadjusted capitation payment rewards a health plan for enrolling healthy people and avoiding the sick — “cream-skimming.” Risk-adjusted capitation counters this by paying more for enrollees predicted to cost more, using either simple cell-based rate tables or full regression models.

The US Medicare progression

From the Adjusted Average Per Capita Cost (AAPCC), to Diagnostic Cost Groups and Ambulatory Care Groups (DCG/ACG), to the CMS Hierarchical Condition Categories (CMS-HCC) model used since 2004 — now covering 100% of the risk-adjusted payment.

US Medicaid uses a parallel system, the Chronic Illness and Disability Payment System (CDPS), adapted for its typically younger, lower-income population.

A real limit: these models can only adjust for what claims actually record, so socioeconomic and clinical detail missing from billing data stays invisible to them, and even the best model predicts individual future spending poorly — prospective R-squared values are generally under 20%, with an estimated ceiling near 15% for total expenditure. Real cream-skimming incentives can therefore survive risk adjustment intact, which is why many systems blend in **partial capitation** — some payment based on actual services used — to soften both the selection incentive and the incentive to underserve sicker enrollees.

Risk Adjustment in Social-Insurance Systems

Bismarck-type systems, where residents choose among competing sickness funds, need risk adjustment for the same reason — without it, funds could profit simply by attracting healthier members.

The Netherlands

An age-and-sex model from 1992, expanded to Pharmacy-based Cost Groups (PCGs) in 2002, then Diagnostic Cost Groups (DCGs) from 2004

Germany

A simple cell-based model from 1994, replaced by a regression-based Risk Structure Equalization scheme since 2009 — expanded from 80 morbidity categories to nearly all 360 disease groups by 2021

Evaluating a Policy Change Without a Trial

Organizational and policy changes are rarely randomized, so HSR leans on the time-series design: repeated observations before a change, the change itself, then repeated observations after — letting a “segmented” or piecemeal regression estimate whether the trend itself shifted, not just the average level.

North Carolina perinatal care

A regionalized perinatal care program was evaluated this way, tracking outcome trends before and after its introduction

Medicaid prescription limits

A cap on reimbursed prescriptions was evaluated the same way, watching for a shift in drug-use trends after the restriction took effect

A trend break right after a policy change is not automatically the policy’s effect — something else could have shifted at the same time (the “piecemeal reform” problem again). A comparison or control series, exposed to the same contemporaneous influences but not the policy itself, is what actually strengthens the causal claim.

Because the outcomes involved are often counts observed repeatedly over time, generalized linear models, generalized estimating equations (GEEs), and Poisson regression are the field’s standard statistical tools for this kind of evaluation.

Worked Example: Reference Pricing for Drugs

Reference pricing caps reimbursement for a class of drugs at the price of its cheapest equally effective option, pushing patients and prescribers toward it.

British Columbia, ACE inhibitors

Reference pricing for this drug class was evaluated using the same time-series and regression methods as other policy changes

Germany, 1992–1993

Reference-based drug prices were introduced at the same time as fixed drug budgets and German reunification — three major changes at once, making the reference-pricing effect alone very hard to isolate

The German case is a textbook illustration of the “piecemeal reform” problem raised earlier: a real policy rarely changes just one thing at a time.

PROCESS OF CARE

When Ordering the Test Pays the Orderer

Self-referral — when the same clinician who orders an imaging test also owns or profits from the equipment — is a well-documented driver of small-area variation: self-referring physicians order X-rays about **four times** as often as those who refer elsewhere. German studies found a similar pattern: a factor of 2.7 for chronic-pain X-rays, and 3.0 for abdominal ultrasound.

Germany performed 1,655 X-rays per 1,000 inhabitants in 1997 — roughly double most other European countries. The stakes are not just financial: about 70% of the population's effective radiation dose comes from just 10% of CT exams, contributing an estimated 2,000 incident cancers a year in Germany alone.

Self-referral and fee-for-service payment reinforce each other: FFS already rewards doing more, and self-referral adds a second, more direct incentive on top of it — the same clinician profits twice from ordering the same extra test.

Worked Example: Radical Prostatectomy, Trial vs. Real World

Outcomes reported in randomized trials of radical prostatectomy were consistently better than outcomes recorded in routine Medicare claims for the same procedure.

Outcome	RCT (Efficacy)	Medicare (Effectiveness)
Cancer recurrence	3%	6%
Incontinence	27%	47%
Erectile dysfunction	60%	89%

Every outcome was worse outside the trial — the gap between what a treatment can achieve under ideal conditions and what it actually achieves in everyday practice.

PROCESS OF CARE, 2 OF 12

Is the Care Even Appropriate?

The RAND/UCLA method uses an expert panel and a structured (Delphi) process to rate whether a procedure was clinically appropriate for the patient who received it.

A historical warning: New York tonsillectomy screening, 1920s

Of 1,000 schoolchildren, doctors recommended tonsillectomy for 611. The remaining children were re-examined by a new panel, which recommended surgery for many of the rest — and this repeated four times. After four rounds of re-examination, only 65 of the original 1,000 children still had tonsils doctors were willing to leave alone.

77%

Coronary angiographies rated appropriate

76%

Endoscopies rated appropriate

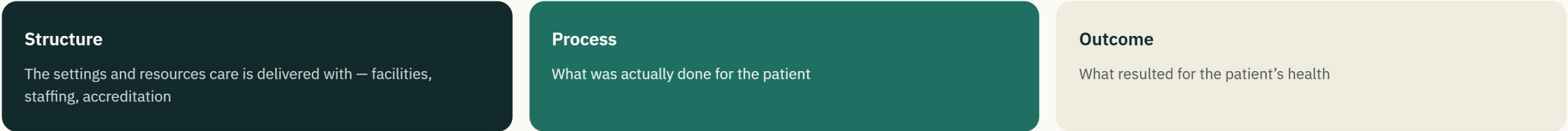
36%

Carotid endarterectomies rated appropriate

The stakes of inappropriate care are real: one congestive heart failure study found 32% mortality among patients given improper treatment, versus 9% among those properly treated.

Donabedian’s Three Ways to Judge Quality

Avedis Donabedian’s enduring framework splits quality measurement into three linked categories.



The three are assumed to be causally linked — good structure enables good process, which produces good outcomes. But before judging each one’s strengths and blind spots, there is a prior question worth asking: quality according to whom?

Quality Depends on Who Is Looking

Blumenthal’s framework identifies four stakeholders who each judge “quality” by different criteria — two can reach opposite verdicts on the very same care without either being simply wrong.

Clinicians

Technical excellence, and the quality of the clinician-patient interaction

Health plans

Population health, and organizational functioning

Purchasers

Price, and the overall effectiveness of the care purchased

Patients

Function, morbidity, mortality, quality of life, and satisfaction

Keep these four viewpoints in mind through the next three slides, which look at structure, process, and outcome measures in turn — each will look more or less convincing depending on whose definition of quality is being applied.

Structural Indicators: Easy to Count, Imperfect Proxies

Accreditation

Bodies like the Joint Commission (JCAHO) and the National Committee for Quality Assurance (NCQA) certify that a facility or plan meets baseline structural standards

Patient volume

Studies by Luft, Hannan, Phibbs, and Thiemann and colleagues repeatedly found that hospitals doing more of a procedure tend to have better outcomes — “practice makes perfect”

But structure is not destiny: even specialist care is no guarantee of better process — some studies have found cardiologists undertreating heart attack patients relative to evidence-based guidelines, despite their specialized training.

Process Indicators: Concrete, But Contested

Process measures check whether a specific, evidence-backed action was taken — for example, giving antibiotics within eight hours for pneumonia, or a beta-blocker after a heart attack.

Why providers like them

They describe something clinicians can directly control, and improve on, without waiting years for an outcome to be observed

Their limits

A single process step is not always a strong predictor of the outcome that matters, and experts often struggle to reach consensus on which steps deserve to be measured

Outcome Indicators: The “Five Ds”



Outcomes are what patients ultimately care about — but comparing them fairly across providers requires case-mix adjustment for how sick patients were to begin with, which is difficult to do well.

Poorly designed outcome measurement can also invite gaming: providers may avoid the sickest patients, or discharge dying patients elsewhere, purely to protect their reported statistics. Case-mix adjustment and risk adjustment differ mainly in their data source — clinical detail versus administrative claims — but both aim at the same goal.

Outcomes are also just hard to use as a ranking tool: differences can take years to show up, may owe more to patient mix or plain chance than to the provider, and a rare outcome like mortality gives most providers too few events for the comparison to have real statistical power.

HEDIS: A Standard Scorecard for Health Plans

The Health Employer Data and Information Set (HEDIS), designed by the NCQA and introduced in 1993, then revised in 1995 and 1997, gave US employers and consumers a standardized way to compare health plans on the same set of quality measures.

Standardization was the innovation: before HEDIS, health plans could each report quality however flattered them most, making real comparison nearly impossible.

PROCESS OF CARE, 9 OF 12

The UK's Hospital League Tables

Dr. Foster's hospital guide, built on Hospital Episode Statistics (HES), rates hospitals across five categories including standardized mortality ratios (SMRs). The Care Quality Commission now assigns hospitals to one of four overall rating categories.

Critics point to real weaknesses: inconsistent control for case mix and comorbidity, gaps and errors inherent to HES administrative data, an over-reliance on a patient's primary diagnosis alone, and the fact that deaths occurring after discharge — outside the hospital — are missed entirely.

PROCESS OF CARE, 10 OF 12

Hospital Rankings, US and German Style

United States

U.S. News & World Report’s “America’s Best Hospitals” (with RTI International) scores structure, process, and outcome together, adjusting severity using APR-DRGs; a companion “Best Children’s Hospitals” ranking follows the same approach

Germany

No single official overall hospital ranking exists; FOCUS magazine instead publishes a widely read clinic comparison — its 2022 list covered around 1,700 clinics across many specialties

Taking the Rankings Down to the Individual Physician

The American Medical Association defines physician profiling as comparing a doctor's practice patterns to those of their peers, often as a ratio of observed to expected events — the same logic as a standardized mortality ratio, applied to one provider.

Massachusetts, 1996

Launched the first public physician-profiling program in the US

Germany, since the 1980s

Benchmark-based profiling of physicians has a long history, reformed most recently by the 2017 GKV Health Care Strengthening Act

OUTCOMES OF CARE

What Did All of This Actually Achieve?

Donabedian's outcome category returns as its own field of study: measuring health status changes at the population level, tracked through indicator sets like the CDC's original 18 leading health indicators from 1991, since expanded into the 23 leading indicators of Healthy People 2030.

This section moves from population vital statistics down to the individual patient's own reported health, then back out to whole health systems compared side by side.

Specific or Generic: Two Families of Patient Measures

Specific measures

Tailored to a disease, organ system, organ, function, or problem — sensitive to change within that one condition, but not comparable across different conditions

Generic measures

Cover overall health status in a way that applies across any condition or population — enabling comparison, at some cost in sensitivity to any one disease's specific concerns

Many studies use both together: a disease-specific instrument to capture what matters most to that patient group, and a generic instrument to allow comparison against other conditions and the general population.

The Generic Instrument Family

SF-36

36 items across eight health concepts, rolled up into Physical and Mental Health summary scores — one of the most widely used instruments worldwide

SIP & QWB

The Sickness Impact Profile and the Quality of Well-Being scale — earlier generic instruments with their own scoring approaches

EQ-5D

Rates 5 dimensions of health at 3 (or, in a newer version, 5) severity levels each; child (EQ-5D-Y) and well-being (5-HWB) variants now exist too

Each instrument makes its own trade-off between depth of detail and ease of completion — a major reason so many different ones remain in active use.

Life Expectancy, Adjusted for Health

Standard life tables (built from either a real cohort followed over time, or a hypothetical period cohort) give raw life expectancy. Two further adjustments weight those years by health rather than counting them equally:

Sullivan’s method
Combines life-table data with disability prevalence to estimate disability-free life expectancy

DALY, the “health gap”
The disability-adjusted life year measures the gap between a population’s actual health and an ideal in which everyone lives to old age free of disease and disability

The WHO’s disability-adjusted life expectancy (DALE) report drew significant controversy over its methods and country rankings — a preview of the same criticisms leveled at cross-country health-system rankings generally, covered later in this section.

OUTCOMES OF CARE, 4 OF 11

QALYs and DALYs Answer Different Questions

QALY

Years **gained** from a treatment or program, weighted up from 0. Built from individual preferences, and used mainly in cost-utility analysis to compare treatment choices.

DALY

Years **lost** to a disease relative to an ideal, weighted down from an ideal life span. Built from population disease burden, and used mainly to compare the overall burden of different diseases.

Put simply: a QALY asks how much a specific intervention can add for one patient or population; a DALY asks how much a disease is already taking away, at the scale of an entire population.

What these summary measures cannot decide for us

Both have drawn ethical criticism. Chasing the lowest cost per QALY can conflict with equity, and because older people and people with disabilities have fewer potential life-years left to gain, the metric can systematically disadvantage them. DALYs inherit related problems, since they too depend on utility-weighting and discounting choices that are ultimately value judgments, not purely technical ones.

OUTCOMES OF CARE, 5 OF 11

Patient Experience Is Its Own Outcome

Beyond clinical results, how care actually felt to the patient has become a growing field in its own right. The OECD's PaRIS initiative (Patient-Reported Indicator Surveys) is building standardized, internationally comparable measures of patient-reported outcomes and experience.

An international study of primary-care priorities spanning the UK, Norway, Sweden, Denmark, the Netherlands, Germany, Portugal, and Israel found that what patients want from general practice care is broadly shared across very different health systems — even when how that care is financed and organized differs enormously.

Three Levels of Efficiency

Technical

Getting the maximum output from a given set of inputs

Productive

Producing a given output at the lowest possible cost

Allocative

Directing resources to the mix of services that yields the most total benefit

Allocative efficiency is judged against the Pareto criterion — a reallocation is an improvement if it can make someone better off without making anyone worse off. Answering allocative questions usually calls for cost-benefit analysis; cost-effectiveness analysis is better suited to narrower technical and productive questions within a fixed budget.

OUTCOMES OF CARE, 7 OF 11

Ranking Whole Health Systems

Whole-system outcomes can be compared cross-sectionally, at a single point in time, or longitudinally, tracking change over years. The WHO’s 2000 World Health Report attempted a single overall country ranking of health system performance — and drew heavy, lasting criticism over its methods and assumptions.

Global Burden of Disease

A comprehensive, continuously updated accounting of disease and injury burden across countries

Effective Coverage Index

Weighs service coverage by the quality of care actually delivered, not just access alone

Healthcare Access & Quality Index

Ranks countries on mortality from causes that good health care should be able to prevent

OUTCOMES OF CARE, 8 OF 11

Deaths That Shouldn't Have Happened

A joint 2019 OECD/Eurostat framework splits “avoidable mortality” into two distinct causes — a distinction that points to two very different policy fixes.

Preventable mortality

Deaths avoidable mainly through public health action — prevention, not treatment

Treatable (amenable) mortality

Deaths avoidable mainly through timely, effective health care once disease is present

Treatable mortality in particular has become a widely used proxy for how well a health *care* system itself — as distinct from broader public health — is performing.

OUTCOMES OF CARE, 9 OF 11

A Fair Comparison Needs Many Measures at Once

A comparison of 13 European countries brought together nine very different measures side by side: disease burden (DALYs), health expenditure, health workforce density, hospital beds and their utilization, unmet need, hospital mortality after a heart attack, the Effective Coverage Index, treatable mortality, and the Healthcare Access and Quality Index.

No single measure told a consistent story: Finland and Germany, for example, showed contradictory patterns depending on which measure was used — a country that looked strong on one indicator could look weak on another. The lesson is the same one raised throughout this chapter: any one number, taken alone, can mislead.

A single year's snapshot can also mislead about cause and effect: a country spending heavily might simply have started from higher need or higher baseline mortality, with that very spending already holding an even worse outcome at bay — a pattern only a longitudinal view, like the one on the next slide, can properly untangle.

OUTCOMES OF CARE, 10 OF 11

Does Spending More Actually Save Lives?

Busse and colleagues (2017) tracked amenable mortality against health expenditure over time across Germany and seven comparator countries — Austria, Canada, Denmark, France, the Netherlands, the UK, and the US — from 2000 to 2015.



A longitudinal view like this one links a system's own spending trend to its own mortality trend over time — a more direct test of whether money spent is money well spent than any single-year cross-country snapshot can offer.

WRAPPING UP

One System, Studied From Every Angle

Every tool in this deck answers a piece of the same underlying question: does the system turn its inputs — money, staff, facilities, patient need — into good outcomes, and how would we know?

Structure, process, and outcome measures each look at a different link in that chain. Study designs, data sources, and confounding control determine whether what they find can be trusted. And demand, need, financing, and efficiency together explain why the system delivers what it delivers in the first place.

WRAPPING UP

A Caution About Comparisons

Simple input and output figures remain politically powerful, precisely because they are easy to report — but life expectancy or spending alone never explains *why* one system does better than another. That explanation still requires looking at structure and process, not just the final number.

Worked example: comparing hospital payment across countries

A German policy debate over per-diem versus case-based hospital payment often invoked US diagnosis-related groups (DRGs) as the free-market alternative — overlooking that only about 15% of the US population was actually covered by DRG-based payment at the time, and ignoring how costs could simply shift into ambulatory care instead of disappearing.

International comparisons are only as good as the local context and mechanisms behind the numbers — borrowed statistics, without borrowed context, can support almost any argument.

BOTTOM LINE

Every Number Describes One Link in a Longer Chain

Health services research succeeds when it matches its study design, data source, and outcome measure to the specific question being asked — whether that question is about access, cost, quality, or the health finally produced — and is honest about what that combination can and cannot prove.

No single figure — life expectancy, spending per capita, a hospital ranking — ever tells the whole story on its own. Understanding a health system means following the chain all the way from its inputs to its outcomes, and questioning any number that skips a link.

As health systems face growing cost pressure and rising expectations, the demand for rigorous, policy-relevant health services research keeps growing along with them.

SOURCE & ATTRIBUTION

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